

Your grade: 100%

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Next item →

1. The basic idea of particle swarm optimization is to simulate _____ in nature.

1 / 1 point

- ☐ species evolution
- ☐ natural selection
- ☐ biological competition
- ☒ fish or bird swarm

👏 Nice work

2. What does the particle's position refer to in the particle swarm optimization algorithm?

1 / 1 point

- ☐ Velocity of the evolving candidate solutions
- ☒ Candidate solutions to the optimization problem
- ☐ The global optimal solution for an optimization problem
- ☐ The location of the current best candidate solution in the current iteration

👏 Nice work

3. What is the difference between“[x] = particleswarm (@PSO_PID, nvars, lb, ub, options) ”and“[x] = particleswarm (@PSO_PID, nvars, [], [], options)”?

1 / 1 point

- ☐ The decision variables are different.
- ☒ The lower bound and upper bound of decision variables are different.
- ☐ The objective function is different.
- ☐ The number of decision variables is different.

👏 Nice work

4. In the code '[x] = particleswarm(PSO_PID, nvars, lb, ub, options)', the function uses particle swarm optimization to solve a proportional-integral-derivative (PID) control problem. What does 'PSO_PID' represent in this context?

1 / 1 point

- ☐ Decision variables
- ☐ Constraints
- ☒ Objective function
- ☐ The number of decision variables

👏 Nice work

5. In the context of particle swarm optimization (PSO), on what basis does each particle update its velocity and position?

1 / 1 point

- ☒ The best position of the whole population

👏 Nice work

- ☒ The particle's current position

👏 Nice work

- ☒ The particle's current velocity

👏 Nice work

- ☒ The best position of the current particle

👏 Nice work